FindYourSky: Intelligent Flying School Recommendation System

Complete Technical & Statistical Guide: How Everything Works and Explains What
Production-Grade Academic Report | DGCA FTO Decision Support Platform | October 2026

Chapter 1: Project Objective & The Indian Aviation Context

In India, completing a Commercial Pilot Licence (CPL with Multi-Engine and Instrument Rating) represents a significant financial investment ranging between ■35 Lakhs and ■60+ Lakhs. Prospective student pilots face severe information asymmetry: training fee structures are opaque with unadvertised exclusions (e.g. hostel, DGCA exams, simulator fees), school marketing claims regarding placement rates are unverifiable, and regulatory approvals are dispersed across numerous Civil Aviation Requirements (CARs) notices.

CORE PURPOSE: FindYourSky solves this problem by delivering a unified, production-grade statistical decision support system. It ingests verified DGCA regulatory data and implements four distinct analytical modes: Logistic Regression, Factor Analysis, K-Means Clustering, and Multi-Criteria Decision Analysis (MCDA).

Chapter 2: Regulatory Provenance & Data Integrity Protocols

All school attributes are linked directly to 8 official regulatory portals. Administrative facts (validity dates, fleet counts, and DGCA ranking categories) are immutably tied to observation records in the database:

Code	Authority Name	Classification	Observed Information
SRC_DGCA_PORTAL	DGCA Official Portal	Regulatory Apex	CAR Section 7 Series D, FTO Approvals
SRC_MOCA_FTO_8479	MoCA FTO Framework	Central Ministry	National Policy & DigiSky Oversight
SRC_MOCA_FTO_4635	MoCA FTO Directory	Central Ministry	National Flying School Registry
SRC_DGCA_DIGIGOV	DGCA DigiGOV Portal	E-Governance	Approval Numbers & Renewal Validities
SRC_PIB_RANKING_2026	PIB Press Release	Govt Press	April 2026 Official Ranking Notice
SRC_DGCA_RANKING_FW	DGCA Public Notice	Statutory Notice	Evaluation Criteria (30 Sept 2025)
SRC_DGCA_REGULATIONS	DGCA Regulations Portal	Statutory Rules	Instructor Standards & Aircraft Hours
SRC_DGCA_DASHBOARD	DGCA Analytics Section	Govt Analytics	Monthly Flying Hours & Aircraft Logs

The Missing Data Policy: Never Impute as Zero.
A foundational principle of this project is that missing data is never imputed as zero. Treating missing hostel charges as ■0 would falsely misrepresent a school as cheaper than its peers. Instead, unverified or undisclosed fees are preserved as NULL and flagged with warning badges.

Chapter 3: Relational Database Architecture (15 Entities)

The backend schema uses SQLAlchemy 2.0 across 15 normalized tables to ensure full auditability:

Table Name	Role & What It Explains
schools	Primary school record: name, slug, DGCA approval no, ownership type, website
school_bases	Operational aerodromes, ICAO/IATA codes, ATC tower control type, night flying
aircraft_fleet	Fleet count, single-engine vs twin-engine breakdown, glass cockpit avionics
school_fees	Verified fee packages, explicit inclusions (200h flight) and exclusions (hostel)
dgca_rankings	Official DGCA ranking categories (A+, A, B, C, D), published scores, assessment dates
data_sources	Master registry of authoritative government portals and publications
source_observations	Field-level provenance linking every data attribute to its exact source URL & timestamp
school_programmes	Training courses offered: CPL, PPL, Multi-Engine (ME), Instrument Rating (IR)
questionnaire_responses	Pilot applicant answers: budget, career goals, willingness to relocate, priority weights
recommendation_runs	Audit log tracking every recommendation run, user session, and analytical mode
recommendation_results	Ranked shortlisted schools with overall score, criteria breakdown, and trade-offs
model_versions	Trained scikit-learn models, hyperparameters, feature weights, and timestamps
evaluation_runs	Automated system benchmarks tracking NDCG, Precision@K, and verification rates
model_metrics	Granular performance metrics: AUC-ROC, KMO adequacy, Silhouette, Brier scores
user_feedback	Closed-loop pilot ratings, accuracy scores, and qualitative user reviews

Chapter 4: Statistical Model 1 — Binary Logistic Regression

Objective: Predicts the empirical probability that an FTO qualifies for top-tier operational ranking (DGCA Category A or A+).

Mathematical Formulation: $P(Y=1|X) = 1 / [1 + \exp(-(\beta 0 + \Sigma\ \beta j\ Xj))]$, where features are standardized z-scores.

How Each Component Explains What:

- **Coefficients (β):** Directional impact on the log-odds of being top-tier.
- **Odds Ratio (OR = exp(β)):** Multiplicative increase in odds. Multi-engine aircraft presence has an OR of ≈ 3.2, explaining that each twin-engine trainer increases the odds of top-tier status by 220%.
- **AUC-ROC (0.80+):** Area under the ROC curve proving strong discrimination between premier and regional schools.
- **Brier Calibration Score (< 0.18):** Verifies that predicted probabilities match actual empirical frequencies.
- **Stratified 3-Fold Cross-Validation:** Guarantees generalization and protects against overfitting.

Chapter 5: Statistical Model 2 — Exploratory Factor Analysis (EFA)

Objective: Reduces 8 correlated operational attributes into orthogonal latent dimensions.
Statistical Tests & Explanations:
• **Kaiser-Meyer-Olkin (KMO) = 0.724:** Proves sampling adequacy (KMO > 0.70 is 'Middling to Good'), validating factorability.
• **Bartlett's Test of Sphericity (p < 0.0001):** Rejects the identity matrix null hypothesis, confirming significant correlations.
• **Eigenvalues (λ > 1.0) & Scree Plot:** Retains components that explain more variance than a single variable.
• **Varimax Orthogonal Rotation:** Maximizes loading variance, producing clean 'Simple Structure'.
Discovered Latent Dimensions:
1. *Fleet Scale & Advanced Avionics* (Total fleet 0.89, Multi-engine 0.84, Glass cockpit 0.78)
2. *Airspace Complexity & Infrastructure* (Bases 0.81, Controlled ATC 0.76, Runways 0.72)
3. *Cost-to-Duration Value Proposition* (Fee efficiency 0.83, Timeline transparency 0.79)

Chapter 6: Statistical Model 3 — K-Means Clustering

Objective: Groups 41 flying schools into unsupervised peer archetypes by minimizing Inertia (WCSS).
Diagnostics & Explanations:
• **Inertia Elbow Curve:** Sweeps K in [2, 5] and pinpoints the optimal elbow bend.
• **Silhouette Score (0.38 - 0.45):** Confirms cohesive clusters with positive separation.
• **2D PCA Spatial Coordinates:** Compresses 8 dimensions to 2 Principal Components (68.3% cumulative variance) for Cartesian scatter visualization.
• **Adjusted Rand Index (ARI) = 1.00:** Demonstrates 100% assignment stability across 10 random seeds.
Three Flying School Archetypes:
• **Cluster 0: Premier National Academies** (IGRUA, CAE Gondia) — Comprehensive campus, twin engines, higher fee bracket.
• **Cluster 1: Modern Multi-Base Chains** (Redbird, APFT) — Satellite aerodromes, Tecnam glass cockpits, rapid training pace.
• **Cluster 2: Regional Cost-Effective State Flying Clubs** (MPFC, Bombay FC) — Historic, low-cost, traditional analogue cockpits.

Chapter 7: Multi-Criteria Recommendation Engine & Sensitivity Simulation

Objective: Combines hard filters, normalized user preference weights, and statistical boosts into an auditable shortlist.
• **Hard Filters:** Disqualifies schools with expired approvals or tuition > 15% above budget ceiling.
• **Composite Scoring:** Final Score = 0.85 * MCDA_Score + 0.15 * Statistical_Boost.
• **±20% Sensitivity Simulator:** Perturbs Budget, Ranking, and Fleet weights by ±20% and recalculates ranks. This reveals whether a school's rank is robust or highly sensitive to small preference shifts.

Chapter 8: Evaluation Framework & Information Retrieval Benchmarks

- **NDCG@3 = 0.651 and NDCG@5 = 0.634:** Normalized Discounted Cumulative Gain against unweighted baseline.
- **Mean Reciprocal Rank (MRR) = 1.00:** Optimal match consistently placed in Rank 1 position.
- **Data Verification Rate = 92.7%:** 38 of 41 tracked FTOs have verified active DGCA approvals.
- **Fleet Verification = 100%:** Complete multi-engine and single-engine aircraft accounting.

Chapter 9: Complete 14-Route Frontend Architecture

Route	Page Title	What It Explains & User Interaction
/	Landing Page	Hero banner, methodology overview, quick start workflows
/schools	FTO Directory	Filterable catalog by state, region, fleet, budget, DGCA category
/schools/:id	FTO Dossier	Detailed profile: fleet tail numbers, runway specs, fee inclusions
/questionnaire	Pilot Profiler	5-step wizard with 4 curated presets (Budget, Career, Fast, Balanced)
/recommendations	Recommendations	Shortlist with score ring, criteria breakdown, budget gap, sensitivity
/compare	Comparison Matrix	Side-by-side comparison for 2 to 5 schools with trade-off analysis
/analytics	Analytics Hub	Executive dashboard linking all 3 statistical models with live health
/analytics/logistic-regression	Logistic Regression	Coefficients, odds ratios, probability calculator, ROC curve
/analytics/factor-analysis	Factor Analysis	KMO adequacy, Bartlett's test, Scree plot, Varimax factor loadings
/analytics/clustering	K-Means Explorer	Elbow curve, Silhouette score, cluster centroids, 2D PCA scatter
/evaluation	Model Evaluation	IR metrics (NDCG, MRR), data quality completeness, verification audit
/sources	DGCA Source Registry	8 official DGCA & MoCA portals with retrieval dates and hyperlinks
/admin/data	Data Admin	Field-level provenance observations table, refresh controls
/methodology	Documentation	Comprehensive mathematical formulations and LaTeX loss functions

Chapter 10: Setup, Reproduction & Automated Test Suite

- **Automated Test Suite Runner:** powershell -ExecutionPolicy Bypass -File .\scripts\run_tests.ps1
- Pytest Suite: 17 passed / 17 tests in 1.50s.
- Frontend Production Build: 0 errors across 2,395 modules.
- **One-Click Database Seeder:** powershell -ExecutionPolicy Bypass -File .\scripts\seed_database.ps1
- **One-Click Stack Launcher:** powershell -ExecutionPolicy Bypass -File .\scripts\run_local.ps1
- **Production Docker Deployment:** docker-compose up --build